

Project Proposal: Automated Data Quality & Metadata Management Platform

Executive Summary

Automated Data Quality and Metadata Management Platform. focus on the data layer, the platform addresses high-value enterprise challenges through automated profiling, intelligent recommendations, and a Human-in-the-Loop (HITL) governance framework.

1. The Problem Statement

- **Pervasive "Dirty Data":** Enterprise pipelines frequently suffer from missing values, duplicate entries, schema drifts, and invalid formats, leading to unreliable analytics and poor decision-making.
- **Siloed Metadata:** Metadata (data context, schemas, data types) is often disconnected from the underlying raw data, making governance, lineage tracking, and automated fixes difficult to implement.
- **Risk of Unsupervised Automation:** Purely automated data remediation without human review introduces operational risks, such as accidentally overwriting valid business logic or corrupting production tables.

2. Our Proposed Solution

We propose an **AI-assisted Data Quality & Metadata Management System** that bridges the gap between raw data analysis and metadata governance.

Key Pillars of the Solution:

- **Multi-Dimensional Data Profiling:** Automated inspection across core data quality dimensions (completeness, uniqueness, accuracy, consistency, and validity).
- **Metadata-Aware Analysis:** Leveraging metadata (schema, column types, relationships) alongside raw data to contextualize issues and generate smarter fixes.
- **Actionable Recommendation Engine:** Rather than merely flagging errors, the system generates actionable remediation strategies (e.g., schema updates, data imputation, type casting).
- **Human-in-the-Loop (HITL) Governance:** A dedicated review layer allowing Data Stewards or Engineers to approve, adjust, or reject proposed fixes before execution, ensuring high system reliability and safety.

3. Application Workflow (System Flow)

Step 1: Data & Metadata Ingestion

- **Inputs:** Connectors pull from relational databases (e.g., PostgreSQL, MySQL), Data Lakes, or static files (CSV, Parquet).
- **Extraction:** The system fetches two layers of information simultaneously:
 - **Raw Data:** Rows, records, and values.
 - **Metadata:** Schemas, data types, table relationships, and column tags.

Step 2: Automated Profiling & Inspection

- **Data Quality Checks:** Scans for missing values, duplicates, format inconsistencies, and statistical outliers.
- **Metadata Checks:** Detects schema drifts, missing documentation/tags, and type conflicts.

Step 3: Intelligent Recommendation Engine

The engine analyzes detected anomalies and formulates contextual fixes:

- *Data Level Example:* Flags 12% missing values in `customer_age` and recommends filling missing entries using median imputation.
- *Metadata Level Example:* Identifies `created_at` stored as a string and recommends casting it to `Timestamp` while updating the metadata catalog.

Step 4: Human-in-the-Loop (HITL) Dashboard

Recommendations are presented in a clean user interface for review:

- **Approve:** Queues the recommendation for automated execution.
- **Modify:** Allows the administrator to adjust parameters (e.g., change imputation strategy).
- **Reject:** Dismisses the recommendation with an optional reason logged for model tuning.

Step 5: Execution & Remediation

Upon approval, the system generates and runs the required scripts or models or pipeline transformations to apply fixes directly to the target environment or staging layer.

Step 6: Audit Logging & Metrics Reporting

- **Audit Trail:** Logs every action (who approved/rejected what change and when) for compliance.

- **Quality Scorecard:** Dynamically calculates and displays the system-wide Data Quality Score improvement (e.g., Quality Score increased from 71% to 95%).